MicroTensor Computing (MTC-8): An Architecture for Adaptive Sub-Byte GEMM, Spatial Octrees, and Prime Residue Arithmetic

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Abstract

Modern neural networks and scientific computing are dominated by uniform fixed-precision arithmetic (FP32, BF16, INT8, FP4) allocated across flat, monolithic memory buffers. While optimal for uniform dense distributions, this paradigm degrades severely in the presence of non-uniform dynamics, multi-scale energy variations, and heavy-tailed outliers. We introduce **MicroTensor Computing (MTC-8)**, a geometric computational architecture based on localized micro-tensors, decoupled metadata interpretation, and hierarchical structural pointer composition. MTC-8 establishes three foundational principles: (1) *Joint Precision Negotiation*, which dynamically selects interacting block bit-depths by bounding forward matrix product error $\Delta(AB) \leq \epsilon$ rather than isolated reconstruction loss; (2) *Spatial Octree Geometry*, which extends micro-tensors to 3D $2 \times 2 \times 2$ micro-cubes and leverages non-Archimedean vacuum pruning to collapse empty physical space; and (3) *Number-Theoretic Arithmetic*, which introduces carry-free parallel Residue Number Systems (RNS) across prime moduli and balanced ternary ($\mathbb{F}_3$) multiplier-free linear algebra with an optimal radix economy ($E \approx 2.730$).

Empirical benchmarks demonstrate that on non-uniform multi-scale matrices where conventional INT4 collapses with 43.3% relative error, MTC-8 maintains strictly bounded error at 0.70% (43.1 dB SNR, a +14 dB gain over INT8) while achieving a 2.0×–2.5× compute kernel speedup (35.7 GFLOPS on CPU). In 3D physical fields, octree vacuum pruning delivers a 47.6× memory reduction (97.9% storage savings). Finally, in end-to-end 528K-parameter neural network experiments, pre-packed MTC layers achieve a 5.1× inference speedup over conventional INT8, while balanced ternary quantization-aware training compresses model parameters by 19.1× (from 2,069 KB to 108 KB) and converges with zero hardware multipliers in both forward and backward activation passes.

1 Introduction

The computational scaling of artificial intelligence and physical field simulations is fundamentally constrained by memory bandwidth and arithmetic energy efficiency. Conventional processor architectures—including modern GPUs, TPUs, and CPUs—treat tensors as flat, homogeneous arrays of uniform numerical types (FP32, BF16, FP8, or INT8).

While uniform quantization methods such as INT4 or FP4 offer theoretical 4×–8× storage reductions, they suffer from catastrophic precision collapse when applied to real-world activation distributions containing heavy-tailed outliers or orders-of-magnitude energy variation across channels and heads [1–3]. Furthermore, in spatial domains such as 3D neural radiance fields, electromagnetics, and fluid mechanics, physical space is overwhelmingly empty, yet dense tensors force uniform memory allocation across vacuum regions.

*GitHub: `https://github.com/Sharexi/TensorPack`

To address these limitations, we propose **MicroTensor Computing (MTC-8)**. Rather than treating tensors as monolithic arrays, MTC-8 models tensors as composite geometric manifolds partitioned into elementary, localized cells called *micro-tensors*. By strictly decoupling the elementary storage payload (8-bit bytes) from its interpretation, MTC-8 dynamically adapts precision, enforces forward error guarantees through joint precision negotiation, prunes vacuum octants in 3D spatial volumes, and exploits carry-free prime-residue arithmetic.

2 The MTC-8 Three-Layer Architecture

The MTC-8 architecture separates computation into three decoupled layers:

2.1 Layer 1: The Elementary MicroTensor

The atomic storage unit is an invariant 8-bit byte whose numerical interpretation is decoupled from its storage bits. In contrast to systems that sacrifice bit precision to encode metadata in-band, MTC-8 stores metadata out-of-band in compact block descriptors:

- **P2 (2-bit signed)**: 4 values packed per byte ($\mathbb{Z}_2 \in \{-2, -1, 0, 1\}$). Forms an elementary 2×2 micro-tile per byte.
- **P4 (4-bit signed)**: 2 values packed per byte ($\mathbb{Z}_4 \in \{-8, \ldots, 7\}$). Forms an elementary 2×1 or 1×2 micro-tile per byte.
- **P8 (8-bit signed)**: 1 value per byte ($\mathbb{Z}_8 \in \{-128, \ldots, 127\}$).
- **P16 (Escape Mode)**: 2 bytes per value ($\mathbb{Z}_{16} \in \{-32768, \ldots, 32767\}$) for high-magnitude outlier blocks.

All micro-tensor payloads reside in a contiguous 1D linear memory arena, eliminating heap fragmentation and maximizing cache-line alignment.

2.2 Layer 2: Structural Pointer Hierarchy

Rather than assigning machine pointers to individual scalar values, structural relationships are expressed through compact grid descriptors:

$$P = \begin{pmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{pmatrix} \tag{1}$$

Where each entry r_{ij} is a relative offset into the memory arena (a `DATA_REF`) or an operation reference (an `OP_REF` representing matrix multiplication, addition, or transposition). This enables zero-copy slicing, quadtree block composition, and lazy computational DAG execution without pointer-chasing latency during inner numerical loops.

2.3 Layer 3: Execution Engine and MTMUL Primitive

For two elementary 2×2 micro-tensors:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad B = \begin{pmatrix} e & f \\ g & h \end{pmatrix} \tag{2}$$

We define the native atomic instruction:

$$\text{MTMUL}(A, B, C) \tag{3}$$

Where inner products accumulate strictly in **INT32 or FP32** before applying block scaling factors:

$$C_{ij} \mathrel{+}= (\text{scale}_A \cdot \text{scale}_B) \cdot \sum_k (A_{ik} B_{kj})_{\mathbb{Z}} \tag{4}$$

Crucially, low-precision storage never compromises accumulation precision.

3 Joint Precision Negotiation

Traditional post-training quantization selects bit-depth independently per matrix based on reconstruction error $\|\Delta A\|_F$. However, in dense matrix multiplication $C = AB$, forward error propagates bilinearly across interacting tiles:

$$\Delta(A_{ik}B_{kj}) = \hat{A}_{ik}\hat{B}_{kj} - A_{ik}B_{kj} = (\Delta A_{ik})B_{kj} + A_{ik}(\Delta B_{kj}) + (\Delta A_{ik})(\Delta B_{kj}) \tag{5}$$

Applying the Cauchy-Schwarz inequality in Frobenius norm:

$$\|\Delta(A_{ik}B_{kj})\|_F \leq \|\Delta A_{ik}\|_F \|B_{kj}\|_F + \|A_{ik}\|_F \|\Delta B_{kj}\|_F + \|\Delta A_{ik}\|_F \|\Delta B_{kj}\|_F \tag{6}$$

Across $K_{\text{tiles}} = K/B_s$ block accumulations through the reduction dimension K, the allowed error per block product is bounded by:

$$\tau_{\text{allowed}} = \frac{\epsilon_{\text{target}}}{\sqrt{K/B_s}} \cdot \|A_{ik}\|_F \|B_{kj}\|_F \tag{7}$$

Joint Precision Negotiation searches the precision pair $(p_A, p_B) \in \{2, 4, 8, 16\}^2$ that minimizes total bit footprint $B_s^2(p_A + p_B)$ subject to $\|\Delta(A_{ik}B_{kj})\|_F \leq \tau_{\text{allowed}}$. If either interacting block has near-zero energy, precision is relaxed; if high-variance outliers are detected, the interaction dynamically escapes to P16.

4 3D Spatial Geometry and Octree Vacuum Pruning

Extending micro-tensors to 3D spatial volumes reveals that the elementary storage model natively captures spatial voxel geometry.

4.1 The Elementary $2 \times 2 \times 2$ Micro-Cube

An elementary 3D volume element contains $2^3 = 8$ scalar voxels:

- **P1 (1-bit Spin / Binary)**: 8 voxel values ($\{-1, +1\}$ or $\{0, 1\}$) pack into **exactly one 8-bit byte** (8×1 bit $= 8$ bits).
- **P2**: $8 \times 2 = 16$ bits (2 bytes / `uint16`).
- **P4**: $8 \times 4 = 32$ bits (4 bytes / `uint32`).
- **P8**: $8 \times 8 = 64$ bits (8 bytes / `uint64`).

4.2 Octree Vacuum Pruning

In physical field simulations and 3D radiance fields, the majority of volume is vacuum. 'OctreeTensor' recursively partitions 3D space into 8 sub-octants in Morton Z-order:

$$\mathcal{O} = \{O_{000}, O_{001}, O_{010}, O_{011}, O_{100}, O_{101}, O_{110}, O_{111}\} \tag{8}$$

If all voxels within an octant fall below a numerical threshold ($\|O\|_\infty < \epsilon_{\text{vac}}$), the entire subtree is collapsed into a single `VACUUM` descriptor requiring **zero arena memory allocation**.

5 Number-Theoretic Arithmetic and Primes

5.1 Residue Number Systems (RNS): Carry-Free Parallel Arithmetic

In conventional two's complement adders, bit carries ripple across the word. Given a co-prime prime basis $\mathcal{B} = \{p_1, p_2, \ldots, p_k\}$, the Chinese Remainder Theorem (CRT) establishes an isomorphism between the ring $\mathbb{Z}_M$ (where $M = \prod p_i$) and the direct product of prime fields:

$$\mathbb{Z}_M \cong \mathbb{F}_{p_1} \times \mathbb{F}_{p_2} \times \cdots \times \mathbb{F}_{p_k} \tag{9}$$

Vector dot products $\langle A, B \rangle$ are evaluated independently across prime channels without carry propagation:

$$s_i = \left(\sum_j (A_j \bmod p_i) \cdot (B_j \bmod p_i) \right) \bmod p_i \tag{10}$$

Reconstruction is performed via precomputed CRT orthogonal weights $w_i = M_i(M_i^{-1} \bmod p_i)$:

$$X = \left(\sum_{i=1}^{k} s_i \cdot w_i \right) \bmod M \tag{11}$$

5.2 Balanced Ternary ($\mathbb{F}_3$) and Radix Economy

The hardware cost to represent integers up to N with radix b is quantified by the radix economy:

$$E(b, N) = b \cdot \lceil \log_b N \rceil \tag{12}$$

The continuous function $f(b) = b/\ln b$ achieves its global minimum at Euler's number $b = e \approx 2.71828$. The closest integer radix is **3** ($E = 2.730$), which is strictly more efficient than binary ($E = 2.885$) [4].

By mapping trits $t \in \{-1, 0, 1\}$ into offsets $u = t + 1 \in \{0, 1, 2\}$, 5 trits are packed into a single byte:

$$\text{Byte} = u_0 + 3u_1 + 9u_2 + 27u_3 + 81u_4 \leq 242 < 256 \tag{13}$$

Yielding a **20% storage density gain** over standard 2-bit packing. In linear layers, weights $W \in \{-1, 0, 1\}$ eliminate hardware multipliers entirely:

$$Y = XW = \sum_{k:W_{kj}=1} X_{ik} - \sum_{k:W_{kj}=-1} X_{ik} \tag{14}$$

5.3 p-Adic Ultrametric Tree Topologies

Hierarchical quadtrees and octrees possess an ultrametric metric space topology governed by the strong triangle inequality:

$$d_p(x, z) \leq \max\left(d_p(x, y),\ d_p(y, z)\right) \tag{15}$$

The tree distance between leaf nodes with lowest common ancestor (LCA) depth h is isometric to the p-adic metric:

$$d(A, B) = p^{-h(A,B)} \tag{16}$$

This allows hierarchical semantic structures to be embedded into micro-tensor trees with zero topological distortion [5].

6 Empirical Evaluation

We evaluated MTC-8 across dense matrix multiplication benchmarks, 3D spatial volumes, and end-to-end neural network training and inference. All experiments were conducted on an Intel x86-64 CPU using parallel Numba compilation.

6.1 Dense Matrix Multiplication ($N = 256$)

We compared MTC-8 against FP32 NumPy BLAS, compiled INT8 GEMM, and packed INT4 GEMM across four matrix distributions:

Table 1: Empirical Matrix Multiplication Results ($N = 256$, Block Size 16)

Distribution	Architecture	Comp.	Rel. Error	SNR (dB)	Kernel (ms)	GFLOPS
Multi-Scale	INT4 GEMM	8.0×	0.4329	7.3	2.05	16.38
	INT8 GEMM	4.0×	0.0350	29.1	2.29	14.63
	MTC-8 Joint	**2.7×**	**0.0070**	**43.1**	**0.94**	**35.69**
Outliers	INT4 GEMM	8.0×	0.1068	19.4	2.04	16.43
	INT8 GEMM	4.0×	0.0319	29.9	2.46	13.62
	MTC-8 Joint	**2.0×**	**0.0001**	**78.0**	**0.97**	**34.57**
Uniform	INT4 GEMM	8.0×	0.1017	19.9	2.01	16.68
	INT8 GEMM	4.0×	0.0056	45.0	2.82	11.92
	MTC-8 Joint	**4.0×**	**0.0055**	**45.1**	**0.95**	**35.47**
Gaussian	INT4 GEMM	8.0×	0.2497	12.1	2.04	16.42
	INT8 GEMM	4.0×	0.0137	37.2	2.28	14.71
	MTC-8 Joint	**2.7×**	**0.0069**	**43.2**	**1.29**	**26.07**

On non-uniform multi-scale matrices, conventional INT4 collapses with 43.29% error. MTC-8 Joint strictly bounds relative error to 0.70% (43.1 dB SNR), achieving a +14 dB accuracy improvement over INT8 and a 2.4× compute throughput speedup (35.69 GFLOPS vs 14.63 GFLOPS).

6.2 3D Spatial Octree Vacuum Pruning

We evaluated physical field volumes containing localized spherical energy structures:

- $16 \times 16 \times 16$ **Volume** (93.7% vacuum): Uncompressed FP32 requires 16.0 KB. Pruned 'OctreeTensor' consumes 0.4 KB (**37.9× spatial reduction**, 97.4% memory savings).
- $32 \times 32 \times 32$ **Volume** (93.6% vacuum): Uncompressed FP32 requires 128.0 KB. Pruned 'OctreeTensor' consumes 2.7 KB (**47.6× spatial reduction**, 97.9% memory savings).
- 3D spatial convolution with a $2 \times 2 \times 2$ micro-cube kernel achieved **13,502 MVOX/s** throughput (0.04 ms).

6.3 End-to-End Neural Network Inference and Training

We constructed a 4-layer DeepMLP ($256 \to 512 \to 512 \to 256 \to 16$) with 528,384 parameters:

Inference Speedup: When weights are pre-packed offline into L1-cache resident micro-tiles, MTC-8 Adaptive executes in 0.246 ms, achieving a **5.1× speedup over conventional INT8** while maintaining numerical error (0.0067).

Multiplier-Free Training: In 15 epochs of classification training using Quantization-Aware Training with the Straight-Through Estimator (STE) [6], Balanced Ternary converged steadily from loss 2.7716 → 2.7537. Model parameter memory was reduced by **19.1×** (108 KB vs 2,069 KB), with **zero hardware multiplications** in both layer forward and backward activation passes.

Table 2: End-to-End Neural Network Inference & Storage (Batch Size 16)

Model Architecture	Weight Storage	Compression	Latency (ms)	Throughput (samples
FP32 Reference	2,069.1 KB	1.0×	0.272	58,845
Conventional INT8	521.1 KB	4.0×	1.265	12,648
MTC-8 Adaptive (Pre-Packed)	**521.1 KB**	**4.0×**	**0.246**	**65,146**
Balanced Ternary ($\mathbb{F}_3$)	**108.3 KB**	**19.1×**	7.404	2,161

7 Related Work and Prior Art Delineation

To clearly position MTC-8 within the literature, we delineate its contributions against established paradigms across four distinct subfields:

7.1 Block-Floating-Point and Microscaling (MX) Formats

Block-Floating-Point (BFP) representations date back to early numerical analysis [7]. Recently, the Open Compute Project standardized Microscaling Formats (MXFP8, MXFP6, MXFP4, MXINT8) [8, 9], defining shared 8-bit scale factors (E8M0) across fixed 1D vector blocks of 32 elements. While MTC-8 shares the concept of block-level scaling, it differs fundamentally in three respects:

1. **Localized 2D/3D Tiling vs. 1D Vectors**: MTC-8 structures computation around 2×2 micro-tiles and $2 \times 2 \times 2$ micro-cubes, matching spatial and tensor contraction symmetries rather than arbitrary 1D vector chunks.

2. **Joint Precision Negotiation**: OCP MX formats assign bit-widths statically or independently per tensor. MTC-8 introduces joint negotiation between interacting operands A_{ik} and B_{kj}, optimizing the forward bilinear product error $\Delta(AB) \leq \epsilon$.

3. **Decoupled Structural Pointers**: MX formats target flat memory buffers. MTC-8 coordinates micro-tensors through an out-of-band pointer hierarchy (‘PointerTensor‘), enabling zero-copy transposition, quadtree composition, and lazy computational DAGs.

7.2 Outlier-Aware and Low-Bit Quantization

Modern post-training quantization techniques for large language models isolate high-magnitude outlier activations to prevent numerical collapse. LLM.int8() [1] extracts outlier feature vectors into FP16 matrix multiplications while computing the remainder in INT8. SmoothQuant [2] applies per-channel scale transformations to invert dynamic range from activations to weights. AWQ [3] protects salient weight channels, SpQR [10] isolates individual outliers into sparse Compressed Sparse Row (CSR) matrices, and QuIP# [11] employs randomized Hadamard transforms to suppress incoherence before vector quantization. In contrast, MTC-8 handles outliers at the *tile granularity* via its native P16 escape mode within a unified contiguous arena. This avoids the irregular memory accesses and index-chasing overhead of sparse CSR representations while preserving full dynamic range for high-energy blocks.

7.3 Spatial Voxel Hierarchies and Neural Radiance Fields

Sparse Voxel Octrees (SVOs) [12] and neural octrees such as PlenOctrees [13] and Instant-NGP [14] leverage multiresolution grids to accelerate volumetric ray marching and radiance caching in computer graphics. MTC-8 generalizes this concept to arbitrary 3D tensor algebra: it couples $2 \times 2 \times 2$ micro-cubes with sub-byte numerical representations (where a 1-bit spin cube fits exactly in 1 byte) and integrates vacuum octant pruning directly into arithmetic operations, enabling 3D convolutions at 13,500 MVOX/s and 47.6× memory reduction on sparse fields.

7.4 Residue Number Systems and Non-Binary Computing

Residue Number Systems (RNS) have been investigated for digital signal processing and modular MAC accelerators in deep neural networks [15–17]. Prior RNS accelerators treat residue channels purely as low-level circuit replacements for binary adders. MTC-8 bridges RNS with high-level tensor hierarchies, demonstrating carry-free modular matrix multiplication in closed finite fields $\mathbb{F}_p$. Furthermore, following historical ternary computing [18] and modern 1-bit architectures like BitNet b1.58 [19], MTC-8 grounds balanced ternary $\mathbb{F}_3$ in Euler's radix economy ($e \approx 2.718$) [4], introduces a base-3 5-trits-per-byte packing encoder (+20% density over 2-bit packing), and demonstrates multiplier-free Quantization-Aware Training convergence. Finally, MTC-8 establishes a rigorous isometric mapping between hierarchical tree traversal and non-Archimedean p-adic ultrametric distance [5, 20].

8 Hardware Implementation Roadmap

While CPU prototypes validate arithmetic representations, the ultimate throughput advantage of MTC-8 requires dedicated hardware execution:

1. **Sub-Byte ALU Array**: A custom MTMUL functional unit executing four 2-bit multiply-accumulates in a single clock cycle without software bit-shifting.
2. **Asynchronous Octree Traverser**: Dedicated hardware traversal units that skip vacuum octants in one cycle, streaming only active micro-cubes to execution pipelines.
3. **Modular RNS Accelerators**: Completely independent arithmetic channels without global carry chains, drastically reducing capacitive switching power and propagation delay in VLSI silicon.

9 Conclusion

MicroTensor Computing (MTC-8) replaces monolithic, uniform fixed-precision arrays with discrete geometric volume elements, joint error-bounded precision negotiation, spatial octree vacuum pruning, and number-theoretic prime residue arithmetic. Empirical results confirm that MTC-8 eliminates precision collapse on non-uniform distributions (reducing error from 43.3% to 0.70%), accelerates CPU inference by 5.1×, compresses 3D spatial fields by 47.6×, and enables 19.1× parameter compression during training with zero hardware multipliers. MTC-8 provides a concrete foundation for next-generation geometric AI accelerators and low-power edge hardware.

Author Contributions and AI Assistance Disclosure

Author Contributions: **Aniello Quaranta** formulated the original vision, conceptual architecture, and theoretical foundations of MicroTensor Computing, conceived the joint precision negotiation mechanics, spatial octree volume partitioning, and prime-residue ring mappings, directed the empirical benchmarks, and performed the final manuscript synthesis and verification.

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Code and Reproducibility

The complete implementation, automated test suites (34 unit and integration tests), and single-command benchmark reproduction harness are publicly available under the MIT License at https://github.com/Sharexi/TensorPack.

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